

Part C

Course title: SC7 Bayes Methods
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Date of this version: 17-04-23

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1. Throughout this question $q(\theta'|\theta) = q(\theta|\theta')$ is a symmetric proposal density.

(a) [15 marks]

- (i) Let $Y \sim p(\cdot|\theta)$, $Y \in \mathcal{Y}$ and let $\theta \in R$ be a parameter with prior $\pi(\theta)$ and, for data $Y = y_{obs}$, posterior $\pi(\theta|y_{obs})$. Explain what it means for $\pi(\theta|y_{obs})$ to be *doubly intractable* and the consequences of this for MCMC targeting $\pi(\theta|y_{obs})$.

Solution: (B, 4) $\pi(\theta|y_{obs}) \propto p(y_{obs}|\theta)\pi(\theta)$. This is DI if either $p(y_{obs}|\theta')/p(y_{obs}|\theta)$ or $\pi(\theta')/\pi(\theta)$ are intractable functions of θ . This may happen if for example the normalisation of $p(y_{obs}|\theta)$ is not available in closed form. MCMC targeting $\pi(\theta|y_{obs})$ does not require the overall normalisation $p(y_{obs})$ for $p(y_{obs}|\theta)\pi(\theta)$ as it cancels in the Metropolis ratio $p(y_{obs}|\theta')\pi(\theta')/p(y_{obs}|\theta)\pi(\theta)$. However if $p(y_{obs}|\theta)$ or $\pi(\theta)$ are intractable then this cannot be evaluated and so MH-MCMC is not possible.

- (ii) Consider the following MCMC-ABC algorithm with summary statistics $S(y) \in R^p$ for $y \in \mathcal{Y}$, distance $d(S, S')$ and threshold δ :

Suppose $X_t = \theta$.

Step 1. Simulate $\theta' \sim q(\cdot|\theta)$ and $y \sim p(\cdot|\theta')$.

Step 2. If $d(S(y), S(y_{obs})) < \delta$ then accept θ' (set $X_{t+1} = \theta'$) with probability

$$\alpha(\theta'|\theta) = \min \{1, \pi(\theta')/\pi(\theta)\}$$

and otherwise reject θ' (set $X_{t+1} = \theta$).

Show this algorithm satisfies detailed balance for $\pi(\theta|d(S(Y), S(y_{obs})) < \delta)$.

Solution: (B,6) Let $E_{y_{obs}}$ be the event that the simulated data is a close match to the data, ie

$$E_{y_{obs}} \Leftrightarrow d(S(Y), S(y_{obs})) < \delta.$$

The probability density to propose θ' is $q(\theta'|\theta)$. The probability to accept it is $p(E_{y_{obs}}|\theta')\alpha(\theta'|\theta)$, so the unnormalised transition density at $\theta' \neq \theta$ is

$$K(\theta, \theta') = q(\theta'|\theta)p(E_{y_{obs}}|\theta')\alpha(\theta'|\theta).$$

We need to check that the algorithm satisfies detailed balance for the target

$$\pi(\theta|E_{y_{obs}}) \propto \pi(\theta)p(E_{y_{obs}}|\theta),$$

ie show

$$\pi(\theta)p(E_{y_{obs}}|\theta)K(\theta, \theta') = \pi(\theta')p(E_{y_{obs}}|\theta')K(\theta', \theta).$$

Canceling the symmetric q factors and taking the LHS,

$$\begin{aligned} LHS &= \pi(\theta)p(E_{y_{obs}}|\theta)p(E_{y_{obs}}|\theta') \min \left\{ 1, \frac{\pi(\theta')}{\pi(\theta)} \right\} \\ &= \pi(\theta')p(E_{y_{obs}}|\theta)p(E_{y_{obs}}|\theta') \min \left\{ \frac{\pi(\theta)}{\pi(\theta')}, 1 \right\} \\ &= \pi(\theta')p(E_{y_{obs}}|\theta')p(E_{y_{obs}}|\theta)\alpha(\theta|\theta') \\ &= RHS. \end{aligned}$$

- (iii) Let $\theta \sim \pi(\cdot|y)$ and $\theta' \sim \pi(\cdot|y_{obs})$. Suppose the following conditions hold.

1. The posterior mean $\mu(S(y)) = E(\theta|Y = y)$ is a linear function of $S(y)$ alone, that is, if $s = S(y)$ and $s_{obs} = S(y_{obs})$ then, for some $\beta \in R^p$,

$$\mu(s) = \mu(s_{obs}) + (s - s_{obs})^T \beta.$$

2. The random variables $\theta - \mu(s)$ and $\theta' - \mu(s_{obs})$ are identically distributed.

Let $\theta_{adj} = \theta - (s - s_{obs})^T \beta$. Show that $\theta_{adj} \sim \pi(\cdot|y_{obs})$ and explain how to use this result to make a regression adjustment to the MCMC-ABC output.

Solution: (S, 5) Since $\theta \sim \theta' + \mu(s) - \mu(s_{obs})$ by (2),

$$\theta - (s - s_{obs})^T \beta \sim \theta' + \mu(s) - \mu(s_{obs}) - (s - s_{obs})^T \beta$$

substituting for the random variable θ . Then since $\mu(s) - \mu(s_{obs}) - (s - s_{obs})^T \beta = 0$ by (1) we have $\theta - (s - s_{obs})^T \beta \sim \theta'$.

For the second part let $\epsilon \sim \theta - \mu(s)$ be the random difference to the “local” mean (ie, at s). Then $\theta = \mu(s) + \epsilon = \mu(s_{obs}) + (s - s_{obs})^T \beta + \epsilon$. Now suppose we run MCMC-ABC and get a sequence of samples $(\theta^{(t)}, s^{(t)})_{t=1}^T$ where $s^{(t)} = S(y^{(t)})$ and $y^{(t)}$ is the y -value simulated at Step 1 in the MCMC-ABC algorithm realising X_t . If we regress $\theta^{(t)} = \alpha + (s^{(t)} - s_{obs})^T \beta + \epsilon^{(t)}$ to get estimates $\hat{\alpha}$ and $\hat{\beta}$ for $\mu(s_{obs})$ and β respectively and set

$$\hat{\theta}_{adj}^{(t)} = \theta^{(t)} - (s^{(t)} - s_{obs})^T \hat{\beta}$$

then $\hat{\theta}_{adj}^{(t)} \sim \pi(\cdot|y_{obs})$ (approximately, as we only have $\hat{\beta}$). (Not required: The purpose in practice is to improve the approximation, allowing us to use larger bandwidth δ and stray further from y_{obs} .)

- (b) [10 marks] Consider the following variant of MCMC. Let $\xi(x|\theta, \theta')$ be a density for real x and let $f(x)$ be a differentiable involution, so $f(f(x)) = x$. Let $f' = df/dx$.

Suppose $X_t = \theta$.

Step 1. Simulate $\theta' \sim q(\cdot|\theta)$ and $x \sim \xi(\cdot|\theta, \theta')$.

Step 2. Accept θ' (set $X_{t+1} = \theta'$) with probability

$$\alpha(\theta'|\theta; x) = \min \left\{ 1, \frac{\pi(\theta'|y_{obs})\xi(f(x)|\theta', \theta)}{\pi(\theta|y_{obs})\xi(x|\theta, \theta')} |f'(x)| \right\}$$

and otherwise reject θ' (set $X_{t+1} = \theta$).

- (i) Show that this algorithm satisfies detailed balance with respect to $\pi(\theta|y_{obs})$. [Hint: What is the marginal probability $\alpha(\theta'|\theta)$ to accept? Also, $f'(x) = 1/f'(f(x))$.]

Solution: (N,6) The answer to the question in the hint is

$$\alpha(\theta'|\theta) = \int \xi(x|\theta, \theta') \alpha(\theta'|\theta; x) dx$$

so detailed balance is satisfied if

$$\pi(\theta|y_{obs})q(\theta'|\theta)\alpha(\theta'|\theta) = \pi(\theta'|y_{obs})q(\theta|\theta')\alpha(\theta|\theta')$$

for all $\theta \neq \theta'$. Cancel the symmetric q both sides and then take the LHS, multiply $\pi(\theta|y_{obs})$ into the integral in $\alpha(\theta'|\theta)$ and pull out $\pi(\theta'|y_{obs})$ then cancel $\pi(\theta'|y_{obs})$ both sides of DB. This leaves, on the RHS,

$$RHS = \int \xi(x|\theta', \theta) \min \left\{ 1, \frac{\pi(\theta|y_{obs})\xi(f(x)|\theta, \theta')}{\pi(\theta'|y_{obs})\xi(x|\theta', \theta)} |f'(x)| \right\} dx$$

and on the LHS,

$$\begin{aligned} LHS &= \int \xi(x|\theta, \theta') \min \left\{ \frac{\pi(\theta|y_{obs})}{\pi(\theta'|y_{obs})}, \frac{\xi(f(x)|\theta', \theta)}{\xi(x|\theta, \theta')} |f'(x)| \right\} dx \\ &= \int \xi(f(x)|\theta', \theta) \min \left\{ 1, \frac{\pi(\theta|y_{obs})\xi(x|\theta, \theta')}{\pi(\theta'|y_{obs})\xi(f(x)|\theta', \theta)} |f'(x)|^{-1} \right\} |f'(x)| dx \\ &= \int \xi(z|\theta', \theta) \min \left\{ 1, \frac{\pi(\theta|y_{obs})\xi(f^{-1}(z)|\theta, \theta')}{\pi(\theta'|y_{obs})\xi(z|\theta', \theta)} |f'(f^{-1}(z))|^{-1} \right\} dz \\ &= \int \xi(z|\theta', \theta) \min \left\{ 1, \frac{\pi(\theta|y_{obs})\xi(f(z)|\theta, \theta')}{\pi(\theta'|y_{obs})\xi(z|\theta', \theta)} |f'(z)| \right\} dz \end{aligned}$$

where we made a change of variables $z = f(x)$ in the third line using $dz = |f'(x)|dx$ and replaced x with $f^{-1}(z)$ wherever it appeared. For the last line, f is an involution so $f^{-1}(z) = f(z)$ and $f'(z) = 1/f'(f(z))$ so LHS=RHS in DB.

(ii) Suppose $p(y|\theta)$ is intractable but we can estimate

$$D(\theta, \theta') = \log (p(y_{obs}|\theta')/p(y_{obs}|\theta))$$

with $\hat{d} \sim N(D(\theta, \theta'), \sigma^2)$. Show that the following algorithm targets $\pi(\theta|y_{obs})$.

Suppose $X_t = \theta$.

Step 1. Simulate $\theta' \sim q(\cdot|\theta)$ and $\hat{d} \sim N(D(\theta, \theta'), \sigma^2)$.

Step 2. Accept θ' (set $X_{t+1} = \theta'$) with probability

$$\alpha(\theta'|\theta; \hat{d}) = \min \left\{ 1, \exp(\hat{d} - \sigma^2/2) \frac{\pi(\theta')}{\pi(\theta)} \right\}$$

and otherwise reject θ' (set $X_{t+1} = \theta$).

[Hint: take $\xi(x; \theta, \theta') = N(x; 0, \sigma^2)$ and $f(x) = \sigma^2 - x$.]

Solution: (N,4) Dropping the hint into the randomised MCMC algorithm we have

$$\begin{aligned} MHR(x) &= \frac{\pi(\theta'|y_{obs})\xi(f(x)|\theta', \theta)}{\pi(\theta|y_{obs})\xi(x|\theta, \theta')} |f'(x)| \\ &= \frac{\pi(\theta')p(y_{obs}|\theta')N(\sigma^2 - x; 0, \sigma^2)}{\pi(\theta)p(y_{obs}|\theta)N(x; 0, \sigma^2)} \\ &= \exp \left(D(\theta, \theta') - \frac{(\sigma^2 - x)^2}{2\sigma^2} + \frac{x^2}{2\sigma^2} \right) \frac{\pi(\theta')}{\pi(\theta)} \\ &= \exp(D(\theta, \theta') + x - \sigma^2/2) \frac{\pi(\theta')}{\pi(\theta)} \end{aligned}$$

and then $D(\theta, \theta') + x \sim \hat{d}$ so we can substitute the random variable $\hat{d}$ and get the stated algorithm (the “penalty method” of Ceperley and Ewing).

2. (a) [7 marks]

- (i) Let S be a sample space and let $\mathcal{S}$ be a σ -field of sets in S . Write down the first three Savage axioms in the form given by de Groot.

Solution: (B,3) In the following events A, B etc are subsets of S and members of $\mathcal{S}$.

Axiom 1. For any two events A and B exactly one of the following relations must hold: $A > B$, $A < B$, $A \sim B$.

Axiom 2. If $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ and $A_i \geq B_i, i = 1, 2$ then $A_1 \cup A_2 \geq B_1 \cup B_2$. If in addition either $A_1 > B_1$ or $A_2 > B_2$ then $A_1 \cup A_2 > B_1 \cup B_2$.

Axiom 3. If $A \in \mathcal{S}$ then $\emptyset \leq A$ with $\emptyset < S$.

- (ii) Suppose $S = [0, 1]$ and $\mathcal{S}$ is the set of all subsets $A \subseteq S$ such that $\int_A dx$ is defined. Suppose our preference is always for the larger interval, so if $A, B \in \mathcal{S}$ then $\int_A dx < \int_B dx$ if and only if $A < B$ with $\int_A dx = \int_B dx$ when $A \sim B$. Also, $\int_\emptyset dx = 0$. Show that these preferences satisfy the first three Savage axioms.

Solution: (S,4) Since $a = \int_A dx$ and $b = \int_B dx$ are two real numbers we must have $a < b$, $a = b$ or $a > b$ so correspondingly $A < B$, $A \sim B$ or $A > B$, so the first axiom is satisfied.

Setting $a_1 = \int_{A_1} dx$ etc, the relations $A_i \geq B_i, i = 1, 2$ imply $a_i \geq b_i$. If $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$ then $\int_{A_1 \cup A_2} dx = a_1 + a_2$ and similarly for the B 's. Since $a_i \geq b_i$ we have $a_1 + a_2 \geq b_1 + b_2$ and hence $A_1 \cup A_2 \geq B_1 \cup B_2$. If in addition $A_1 > B_1$ then $a_1 > b_1$ so there exists $\epsilon > 0$ such that $a_1 - b_1 \geq \epsilon$. Since $a_2 - b_2 \geq 0$ we have $a_1 + a_2 \geq b_1 + b_2 + \epsilon$ so $a_1 + a_2 > b_1 + b_2$ and hence $A_1 \cup A_2 > B_1 \cup B_2$. This shows the second axiom is satisfied.

Finally, taking the measure of the empty set to be zero, since $\int_A dx \geq 0$ we have $\emptyset \leq A$, and since $\int_S dx = 1$ and $0 < 1$, also $\emptyset < S$.

- (b) [8 marks] Let $X_1, X_2, X_3, \dots$ be an infinite exchangeable sequence of binary random variables.

- (i) What does it mean to say that these variables are exchangeable?

Solution: (B,2) For $n = 1, 2, 3, \dots$ let $p(x_1, \dots, x_n) = \Pr\{X_1 = x_1, \dots, X_n = x_n\}$. The sequence is exchangeable if

$$p(x_1, x_2, \dots, x_n) = p(x_{\sigma_1}, x_{\sigma_2}, \dots, x_{\sigma_n})$$

for each $n \geq 1$ and any permutation $\sigma = (\sigma_1, \sigma_2, \dots, \sigma_n)$ of the numbers $(1, 2, \dots, n)$.

- (ii) State de Finetti's theorem for the sequence $X_1, X_2, X_3, \dots$

Solution: (B,2) There exists a distribution function $F(\theta)$ on $[0, 1]$ such that

$$p(x_1, \dots, x_n) = \int_0^1 \left[\prod_{i=1}^n p(x_i | \theta) \right] dF(\theta)$$

where $p(x_i | \theta) = \theta^{x_i} (1 - \theta)^{1-x_i}$. The random variable

$$\Theta = \lim_{N \rightarrow \infty} N^{-1} \sum_{i=1}^N X_i \text{ exists and } F(\theta) = \Pr(\Theta \leq \theta).$$

- (iii) Show that $\Pr(X_1 = x_1, X_2 = x_2, X_3 = x_3) = \Pr(X_{98} = x_1, X_{99} = x_2, X_{100} = x_3)$ for all $(x_1, x_2, x_3) \in \{0, 1\}^3$.

Solution: (S,4) By de Finetti a generative model with $Q \sim F$ realising $Q = q$ and $X_i \sim p(\cdot|q)$ iid exists, so $\Pr(X_1 = x_1, X_2 = x_2, X_3 = x_3|Q = q) = \Pr(X_{98} = x_1, X_{99} = x_2, X_{100} = x_3|Q = q)$. Then

$$\begin{aligned}\Pr(X_1 = x_1, X_2 = x_2, X_3 = x_3) &= E_{Q \sim F}(\Pr(X_1 = x_1, X_2 = x_2, X_3 = x_3|Q)) \\ &= E_{Q \sim F}(\Pr(X_{98} = x_1, X_{99} = x_2, X_{100} = x_3|Q)) \\ &= \Pr(X_{98} = x_1, X_{99} = x_2, X_{100} = x_3).\end{aligned}$$

- (c) [10 marks] For $\theta \in R^p$ and $\psi \in R^q$ let $p_0(\theta) = q_0(\theta)/c_0$, $p_1(\theta, \psi) = q_1(\theta, \psi)/c_1$ and $p_2(\theta, \psi) = q_2(\theta, \psi)/c_2$ be probability densities with intractable normalising constants c_0, c_1 and c_2 . Let $Q(\psi)$ be a probability density which is tractable and normalised. Let $r = c_0/c_2$. The following results are subject to regularity conditions which are omitted.

- (i) Show that

$$r = \frac{E_{(\theta, \psi) \sim p_1}(q_0(\theta)Q(\psi)/q_1(\theta, \psi))}{E_{(\theta, \psi) \sim p_1}(q_2(\theta, \psi)/q_1(\theta, \psi))}$$

Solution: (S,3)

$$\begin{aligned}E_{(\theta, \psi) \sim p_1}(q_0(\theta)Q(\psi)/q_1(\theta, \psi)) &= \int p_1(\theta, \psi)q_0(\theta)Q(\psi)/q_1(\theta, \psi)d\theta d\psi \\ &= \int q_0(\theta)Q(\psi)/c_1 d\theta d\psi \\ &= c_0/c_1\end{aligned}$$

since $\int Q(\psi)d\psi = 1$. Similarly $E_{(\theta, \psi) \sim p_1}(q_2(\theta, \psi)/q_1(\theta, \psi)) = c_2/c_1$ so the ratio gives c_0/c_2 .

- (ii) Let $(\theta^{(t)}, \psi^{(t)}) \sim p_1$ be independent for $t = 1, \dots, T$. Consider the r -estimator

$$\hat{r} = \frac{\sum_{t=1}^T q_0(\theta^{(t)})Q(\psi^{(t)})/q_1(\theta^{(t)}, \psi^{(t)})}{\sum_{t=1}^T q_2(\theta^{(t)}, \psi^{(t)})/q_1(\theta^{(t)}, \psi^{(t)})},$$

and let the relative mean square error be defined as

$$RE(\hat{r}) = \frac{E[(\hat{r} - r)^2]}{r^2},$$

where the expectation is taken over the random samples $(\theta^{(t)}, \psi^{(t)})$, $t = 1, \dots, T$. It may be shown (using the delta-rule) that

$$RE(\hat{r}) = \frac{1}{T} E_{(\theta, \psi) \sim p_1} \left(\frac{(p_0(\theta)Q(\psi) - p_2(\theta, \psi))^2}{p_1(\theta, \psi)^2} \right) + o(T^{-1}).$$

Show that $RE(\hat{r})$ is minimised to leading order in T over densities p_1 by the choice

$$p_1(\theta, \psi) \propto |p_0(\theta)Q(\psi) - p_2(\theta, \psi)|.$$

[Hint: the Cauchy Schwarz inequality is $|\int a(x)b(x)dx|^2 \leq \int |a(x)|^2 dx \int |b(x)|^2 dx$.]

Solution: (N,7)

$$\begin{aligned}
 E_{(\theta, \psi) \sim p_1} \left(\frac{(p_0(\theta)Q(\psi) - p_2(\theta, \psi))^2}{p_1(\theta, \psi)^2} \right) &= \int \frac{(p_0(\theta)Q(\psi) - p_2(\theta, \psi))^2}{p_1(\theta, \psi)} d\theta d\psi \\
 &= \int \left(\frac{p_0(\theta)Q(\psi) - p_2(\theta, \psi)}{\sqrt{p_1(\theta, \psi)}} \right)^2 d\theta d\psi \int (\sqrt{p_1(\theta, \psi)})^2 d\theta d\psi \\
 &\geq \left(\int \left| \frac{p_0(\theta)Q(\psi) - p_2(\theta, \psi)}{\sqrt{p_1(\theta, \psi)}} \right| \sqrt{p_1(\theta, \psi)} d\theta d\psi \right)^2 \\
 &= \left(\int |p_0(\theta)Q(\psi) - p_2(\theta, \psi)| d\theta d\psi \right)^2,
 \end{aligned}$$

with equality holding if we take

$$q_1^{(opt)}(\theta, \psi) = |p_0(\theta)Q(\psi) - p_2(\theta, \psi)|$$

and $p_1^{(opt)} = q_1^{(opt)} / c_1^{(opt)}$ as we now show:

$$\begin{aligned}
 E_{(\theta, \psi) \sim p_1^{(opt)}} \left(\frac{(p_0(\theta)Q(\psi) - p_2(\theta, \psi))^2}{p_1^{(opt)}(\theta, \psi)^2} \right) &= c_1^{(opt)} \int \frac{(p_0(\theta)Q(\psi) - p_2(\theta, \psi))^2}{q_1^{(opt)}(\theta, \psi)} d\theta d\psi \\
 &= c_1^{(opt)} \int |p_0(\theta)Q(\psi) - p_2(\theta, \psi)| d\theta d\psi \\
 &= \left(\int |p_0(\theta)Q(\psi) - p_2(\theta, \psi)| d\theta d\psi \right)^2.
 \end{aligned}$$

The risk is therefore minimised at $p_1 = p_1^{(opt)}$. *Remark (not required): we still have a degree of freedom in Q ; the risk can be further optimised by taking $Q(\psi) = p_2(\psi|\theta)$. See MH Chen and QM Shao Statistica Sinica 7:607-630 (1997).*

3. (a) [13 marks] Let $\alpha > 0$ be a real parameter and H a base distribution on R . Let $\theta = (\theta_1, \dots, \theta_n)$, $\theta \in R^n$. Let $G \sim \Pi(\alpha, H)$ be a Dirichlet Process (DP) and let $\theta_i \sim G$ independently for $i = 1, \dots, n$. Let $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ give the unique values in θ and let $S = (S_1, \dots, S_K)$ partition $[n] = \{1, 2, \dots, n\}$ with $S_k = \{i \in [n] : \theta_i = \theta_k^*\}$ for $k \in [K]$.

Recall the Chinese Restaurant Process (CRP) realising (θ^*, S) :

1. set $K = 1$, $S_1 = \{1\}$ and $S = \{S_1\}$
2. for $i = 1, \dots, n - 1$
with probability $\alpha/(\alpha + i)$ do (a) and otherwise do (b):
(a) set $S_{K+1} = \{i + 1\}$, $S \leftarrow S \cup S_{K+1}$ and $K \leftarrow K + 1$.
(b) simulate $k \sim (|S_1|, \dots, |S_K|)/i$ and set $S_k \leftarrow S_k \cup \{i + 1\}$.
3. simulate $\theta_k^* \sim H$ independently for $k = 1, \dots, K$.

Let $P_{\alpha, [n]}(S)$ give the probability to output partition S for “customer” labels $[n]$.

- (i) Calculate $P_{\alpha, [n]}(S)$ in terms of α , K and the set sizes $(|S_1|, \dots, |S_K|)$.

Solution: (B,6) Each arrival $i = 1, \dots, n - 1$ brings a denominator factor $(\alpha + i)^{-1}$, so the denominator is $\prod_{i=1}^{n-1} (\alpha + i) = \prod_{i=2}^n (\alpha + i - 1)$.

Now look at the numerator. Suppose at the end $S_k = \{i_1, i_2, \dots, i_{n_k}\}$. When i_1 arrived, i was equal $i_1 - 1$ and $k - 1$ tables were occupied, so i_1 creates a new set S_k via option 2a with probability $\alpha/(\alpha + i_1 - 1)$ (and we already accounted for the denominator).

After that, for $j = 2, \dots, n_k$, when i_j arrived $i = i_j - 1$ and set S_k had $j - 1$ elements, so i_j chose option 2b w.p. $(i_j - 1)/(\alpha + i_j - 1)$ and then set S_k w.p. $(j - 1)/(i_j - 1)$, so the factor is $(j - 1)/(\alpha + i_j - 1)$. Including the first arrival in this set, the overall numerator factor from table k is $\alpha(n_k - 1)!$.

If we end up with K tables then there are $K - 1$ events in which a new table is chosen so

$$\begin{aligned} P_{\alpha, [n]}(S) &= \alpha^{K-1} \prod_{k=1}^K (n_k - 1)! \prod_{i=2}^n \frac{1}{\alpha + i - 1} \\ &= \alpha^K \prod_{k=1}^K (n_k - 1)! \prod_{i=1}^n \frac{1}{\alpha + i - 1} \\ &= \frac{\Gamma(\alpha)}{\Gamma(\alpha + n)} \alpha^K \prod_{k=1}^K \Gamma(n_k) \end{aligned}$$

where we extended the product in the denominator down to one in the first step and in the second step we used $\Gamma(x) = (x - 1)\Gamma(x - 1)$ for $x > 1$ to get $\Gamma(\alpha + n) = \Gamma(\alpha) \prod_{i=1}^n (\alpha + i - 1)$. For full marks the first line is fine.

- (ii) Suppose $S \sim P_{\alpha, [n]}$. Take $\sigma \subset [n]$ and let $S^{(\sigma)} = (S_1^{(\sigma)}, \dots, S_{K^{(\sigma)}}^{(\sigma)})$ be the partition we get if we run the CRP to get S and then remove any customers not in σ . Any sets left empty in S are removed, so $1 \leq K^{(\sigma)} \leq K$. Show that $S^{(\sigma)} \sim P_{\alpha, \sigma}$.

Solution: (S, 4) In the algorithm above the “customers” arrive in the order $(1, 2, \dots, n)$. However, $P_{\alpha, [n]}(S)$ does not depend on the order of the arrivals, so we can reorder them so that the customers in σ arrive before those in $[n] \setminus \sigma$. If we stop the process after the customers in σ have arrived we get a realisation $T \sim P_{\alpha, \sigma}$. If we continue the process to realise S and then remove the elements

in the complement of σ from S then we just recover $S^{(\sigma)} = T$, so $S^{(\sigma)} \sim P_{\alpha, \sigma}$.

(iii) Let $n=100$ and $\alpha=2$. Give the probability 98, 99 and 100 are all in different sets.

Solution: (S,3) Exchangeable so same as probability 1, 2 and 3 are in different sets, which is $\alpha^2/(\alpha+1)(\alpha+2) = 1/3$.

(b) [12 marks] Let X be an $m \times n$ binary matrix with $X_{i,j} \sim \text{Bernoulli}(\theta_{i,j})$, $(i,j) \in [m] \times [n]$ all independent. Cluster the $\theta_{i,j}$ -values in different “cells” (i,j) : let $R = \{R_1, \dots, R_K\}$ partition the row labels $[m]$ and for $k \in [K]$ let $C_k = \{C_{k,1}, \dots, C_{k,L_k}\}$ partition the column labels $[n]$ in row-block R_k . For $k \in [K]$ and $\ell = 1, \dots, L_k$ let

$$c_{k,\ell} = \{(i,j) \in [m] \times [n] : i \in R_k, j \in C_{k,\ell}\}$$

be the cells in column partition ℓ of row-partition k . Let $\theta_{k,\ell}^* \in [0, 1]$ be the probability parameter for all cells in $c_{k,\ell}$, so that $\theta_{i,j} = \theta_{k,\ell}^*$ for all cells $(i,j) \in c_{k,\ell}$.

(i) Let $C = (C_1, \dots, C_K)$ and $\theta^* = (\theta_{k,\ell}^*)_{k \in [K], \ell \in [L_k]}$. The prior for the row partition is $P_{\alpha, [m]}(R)$. The column partitions have independent priors $P_{\beta, [n]}(C_k)$ and the probability parameters have independent priors $h(\theta_{k,\ell}^*) = \text{Beta}(\theta_{k,\ell}^*; u, v)$.

Give the posterior $\pi(R, C, \theta^* | X)$ in terms of the priors and observation model (up to a constant not depending on R, C, θ^*).

Solution: (N,4) Let $N_{k,\ell} = |c_{k,\ell}|$ and $n_{k,\ell} = \sum_{(i,j) \in c_{k,\ell}} X_{i,j}$. The posterior is

$$\pi(R, C, \theta^* | X) \propto P_{\alpha, [m]}(R) \prod_{k=1}^K \left[P_{\beta, [n]}(C_k) \prod_{\ell=1}^{L_k} h(\theta_{k,\ell}^*) (\theta_{k,\ell}^*)^{n_{k,\ell}} (1 - \theta_{k,\ell}^*)^{N_{k,\ell} - n_{k,\ell}} \right]$$

(ii) Calculate $\pi(R, C | X)$ (up to a constant) by integrating $\pi(R, C, \theta^* | X)$ over θ^* .

Solution: (N,3)

$$\begin{aligned} \pi(R, C, \theta^* | X) &= \int \pi(R, C, \theta^* | X) d\theta^* \\ &\propto P_{\alpha, [m]}(R) \prod_{k=1}^K \left[P_{\beta, [n]}(C_k) \prod_{\ell=1}^{L_k} \int \frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)} (\theta_{k,\ell}^*)^{n_{k,\ell}+u-1} (1 - \theta_{k,\ell}^*)^{N_{k,\ell}-n_{k,\ell}+v-1} d\theta_{k,\ell}^* \right] \\ &= P_{\alpha, [m]}(R) \left(\frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)} \right)^{\sum_{k=1}^K L_k} \prod_{k=1}^K \left[P_{\beta, [n]}(C_k) \prod_{\ell=1}^{L_k} \frac{\Gamma(u+n_{k,\ell})\Gamma(v+N_{k,\ell}-n_{k,\ell})}{\Gamma(u+v+N_{k,\ell})} \right]. \end{aligned}$$

Notice we need to keep the normalising constant for the Beta distribution, as it appears in $\sum_{k=1}^K L_k$ factors, and this depends on R and C .

(iii) Let $Q \sim \Pi(a, \Pi(b, F))$ be a DP with a base distribution $\Pi(b, F)$ which is itself a DP with base distribution F . Let $G_i \sim Q$ independently for $i \in [m]$ and $G = (G_1, \dots, G_m)$. Let $G^* = (G_1^*, \dots, G_K^*)$ give the unique G_i^* 's in G and let S be the corresponding partition of $[m]$. For $k \in [K]$ let $p_{k,j} \sim G_k^*$ independently for $j \in [n]$ and $p_k = (p_{k,1}, \dots, p_{k,n})$. Let $p_k^* = (p_{k,1}^*, \dots, p_{k,L_k}^*)$ give the unique $p_{k,j}$'s in p_k and let D_k give the corresponding partition of $[n]$. Let $D = (D_1, \dots, D_K)$ and

$p^* = (p_1^*, \dots, p_K^*)$.

Show that (S, D, p^*) has the same distribution as (R, C, θ^*) for some suitable a, b and F which you should give.

Solution: (N,5) The prior for (R, C, θ^*) is

$$\pi(R, C, \theta^*) = P_{\alpha, [m]}(R) \prod_{k=1}^K \left[P_{\beta, [n]}(C_k) \prod_{\ell=1}^{L_k} h(\theta_{k, \ell}^*) \right].$$

The atoms of Q are realisations of $\Pi(b, F)$, so each one is a DP. When we map from G to (S, G^*) , the partition $S \sim P_{a, [m]}$ by the result we established in Question 3.a(i). Now $G_k^* \sim \Pi(b, F)$ iid (by the last line of the algorithm given in 3.a) and each one is a DP. Its atoms are samples from F so if $p_{k, j} \sim G_k^*$ independently for $j = 1, \dots, n$ and we map p_k to (p_k^*, D_k) then $D_k \sim P_{b, [n]}$ (by 3.a(i) again). Also, $p_{k, \ell}^* \sim F$ independently for $\ell = 1, \dots, L_k$ (by the algorithm in 3.a again). Let f be the density for F . Gathering the factors,

$$p(S, D, p^*) = P_{a, [m]}(S) \prod_{k=1}^K \left[P_{b, [n]}(D_k) \prod_{\ell=1}^{L_k} f(p_{k, \ell}^*) \right]$$

so we match $\pi(R, C, \theta^*)$ if we take $a = \alpha$, $b = \beta$ and $f(p_{k, \ell}^*) = h(p_{k, \ell}^*)$, that is, $F = \text{Beta}(u, v)$. *Remark (not required): this is similar to, but not exactly the same as, the “Nested Dirichlet Process”, one of many hierarchical DP variants.*